

KENDRIYA VIDYALAYA SANGATHAN, BHOPAL REGION

MARKING SCHEME -PRE-BOARD EXAMINATION (2024-25)

SET NO. 03

Max. Marks: 80

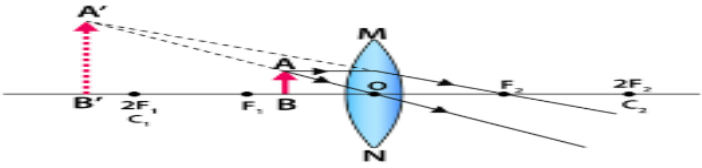
CLASS-X

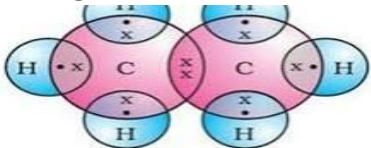
SCIENCE (086)

Time: 3 hours

	SECTION A	
1	(d)	[1]
2	(a)	[1]
3	(c)	[1]
4	(a)	[1]
5	(c)	[1]
6	(c)	[1]
7	(b)	[1]
8	(c)	[1]
9	(c)	[1]
10	(c)	[1]
11	(c)	[1]
12	(d)	[1]
13	(c)	[1]
14	(d)	[1]
15	(d)	[1]
16	(a)	[1]
17	(d)	[1]
18	(d)	[1]
19	(a)	[1]
20	(b)	[1]
	SECTION B	
21	It gets decomposes in light. (0.5), Photochemical/ light decomposition reaction (0.5), $2\text{AgCl} \xrightarrow{\text{light}} 2\text{Ag} + \text{Cl}_2$ (1)	[2]

22	Alveoli(0.5), nephron(0.5), any one function of each (0.5+0.5) OR Function of each enzyme (0.5*4)	[2]
23	(a) responsible for excretion/respiration/nutrition (any one --- 1) (b) leads to female feticide/ disturbs sex ratio/ any other reason--- 1)	[2]
24	(a) Myopia/short sightedness/near sightedness....(0.5) (b) $F=1/P = 1/-4.5 = -0.22$ D----- (1) (c) Concave lens/diverging/ real inverted ---- (0.5)	[2]
25	(i) concentric circles/ drawing them/anticlockwise.... 0.5+0.5 (ii) right hand thumb rule with statement --- 0.5 + 0.5	[2]
26	Bio-degradable & non- biodegradable waste---0.5+0.5, cleaning agents / nutrient recycling/ soil enrichment---one reason ----1 mark	[2]
SECTION C		
27	A Phototropism,shoot is positively phototropic (1) B Movement happens at a point different from the point of touch. Plant cells change shape by changing the amount of water in them, resulting in swelling or shrinking, and therefore in changing shape. (1) C. Growth of pollen tubes towards the ovule is an example of chemotropism whereas bending of shoots towards sunlight is an example of phototropism. (1)	[3]
28	A—ethanoic acid, B--- ester/ ethyl ethanoate. (0.5*2= 1), esterification (0.5)+ correct equation(0.5), saponification reaction(0.5) + correct equation(0.5)	[3]
29	(Any letter which clearly indicated dominant and recessive ears, example, L or E or any other) A. $LL \times ll$ F1----- Ll Ll X Ll 1LL:2Ll:1ll. (2) B. No change in ratio/the ratio of F2 generation will still be 1LL:2Ll:1ll/ ratio will be the same. As the cross is still between a pure dominant and recessive allele/ genes/ traits/characters /as shown in the cross above. (1)	[3]
30	A. Hypermetropia is a defect that causes difficulty in focusing on near objects, with clearer vision observed for distant objects. In Myopia distant objects appear blurry while near objects are seen clearly. (1) B. (i) The image shows a case of hypermetropia. (0.5) (ii) shortening of the eyeball or focal length of the eye lens becomes too long. (0.5) (iii) No the concave lens would diverge the rays coming to the eyeball and will push the image even further, but a convex lens should be used which will help to converge the rays	[3]

	and create the image at the exact place on the retina. (1)	
31	(a) high reactivity/oxidises easily....1 (b) to prevent rusting/ corrosion....1 (c) inter molecular force of attraction is strong than covalent bonds...1	[3]
32	(a) live has 220 V supply but neutral wire has 0 V , high flow of electrons..1 (b) equal voltage distribution , continuous supply without line break.---1 (c) prevents appliances from heavy voltage/current/short circuiting---1 OR Coil with high number of turns of insulated wire---- 0.5, correct diagram---1, use to make electromagnet---- 0.5, any two properties..0.5+0.5	[3]
33	A. heat produced is due to electric effect of current----- 0.5, $H = I^2Rt$ --- 0.5 B. $\eta_f = \frac{2 \times 2}{2} = 100 \times 41000 = 0.4 \frac{2 \times 2}{2} h \frac{2 \times 2}{2} = \frac{2 \times 2}{2} = 500 \times 21000 = 1 \frac{2 \times 2}{2} h$ (1) $\frac{2 \times 2}{2}$ Total energy consumed in 1 day = $\frac{2 \times 2}{2} + \frac{2 \times 2}{2} = 1.4 \frac{2 \times 2}{2} h$ Total energy consumed in 60 days = $1.4 \times 60 = 84 \frac{2 \times 2}{2} h$. Total cost = $84 \times \frac{2 \times 2}{2} \times 6.5 = \frac{2 \times 2}{2} \times 546.00$ (1)	[3]
	SECTION D	
34	a) In chlor-alkali process(0.5), chlorine and hydrogen gases are formed as by-products along with sodium hydroxide. X is chlorine gas, and Y is bleaching powder.(0.5+0.5) Correct equations with balancing-----1.5 mark b) Methanoic acid/formic acid-----0.5, formula--- 0.5, rubbing any base like baking soda/ calamine---- 1 mark OR (i)methane , CH_4 ---0.5 , its structure---1 marks, covalent bonds---0.5 (ii)any one difference....1 mark (iii)acetic acid freezes in winter at room temperature--- 1 mark (iv)Ethene 1 mark	[5]
35	A) For each correct ray diagram --- 1 mark x 2 = 2 B) Ratio of sin I to sin r is always a constat--(1), calculation of RI= 1.5---(2) OR Given: $u = 50 - 12 = -38$ cm----0.5, $v = 88 - 50 = 38$ cm-----0.5 (i) using lens formula, $f = 19$ cm-----1 (ii) using lens formula, $v = \text{infinity}$ -----1 (iii) virtual, erect and magnified image-----1 (iv) correct ray diagram with arrows marked-----2 	[5]

36	(i) Vegetative propagation/ asexual reproduction. (0.5+0.5) (ii) More crops in same time interval, genetically identical, flower fruit faster, no need to depend on pollinators. (1+1) (iii) Cross pollination, the pollen from anther will be transferred the stigma of another plant using agents like wind, water, or insects etc. (0.5+0.5) (iv) There would be minor changes/some variation during the process of copying of the DNA. (1) OR (i) Difference between both terms----0.5+0.5 (ii) Site: ovule, product: zygote/embryo----0.5+0.5 (iii) Diagram with any 4 labeling-----0.5*4= 2 (iv) Ovule to seeds , ovary to fruit-----0.5+0.5	[5]
SECTION E		
37	(i) It will turn milky.-----1 (ii) $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2$ $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$-----1/2 x 2=2 (iii) With excess CO_2, milkiness disappears. $\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{Ca}(\text{HCO}_3)_2$-----1/2 x 2=2 (iv) $\text{C}_2\text{H}_5\text{OH} + \text{Na}_2\text{CO}_3 \rightarrow$ No change -----1 OR i) Methanol is oxidised to methanal in the liver. Methanal reacts rapidly with the components of cells. It causes the protoplasm to coagulate. -----1  Figure 4.6 ii) -----1 iii) a) hexanone b) butanoic acid c) pentanal d) ethanol (0.5 x 4)	[4]
38	1.1) 50% or 1/2-----1 1.2) 22 pairs i.e. 44 autosomes-----1 1.3) Two types of sperms carrying either X or Y chromosome-----1 1.4) Turtle/snail/reptiles/crocodiles----- 0.5+0.5 OR 2.1) F2 / SECOND generation-----1 2.2) Round yellow (9), wrinkled yellow (3), round green (3), and wrinkled green seeded plants (1). Hence, the phenotypic ratio is 9:3:3:1. -----1 2.3) in F1 , only dominant trait expressed like in monohybrid cross, phenotypic	[4]

	and genotypic ratios differ as 3:1 vs 1:2:1. But in F2, effect of recessive trait appeared,-----2	
39	1.1) (a) 2 V -----1 1.2) (C) -----1 1.3) Parallel combination ----1 1.4) (c) $(x + y)^2/xy$ -----1	[4]